

# BuildOrbit — Comprehensive Step-by-Step Testing Guide

This document is engineered to provide Quality Assurance (QA) analysts, project stakeholders, and end-clients with exact, reproducible testing procedures. Following these steps will empirically verify that all critical enterprise features, Role-Based Access Controls (RBAC), and backend data integrations within the BuildOrbit platform are fully functional and production-ready.

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## 1. Environment Preparation & Prerequisites

Before commencing the testing phase, the system must be compiled and populated with deterministic test data.

1. **Initialize the Production Build:** Ensure the Next.js server is running the highly optimized production build, not the development server.

```
npm run build
npm start
```

2. **Seed the Database:** Inject the standard suite of mock enterprise data (users, leads, leave types, and attendance history) into your PostgreSQL instance.

```
npx prisma db push --accept-data-loss
npx prisma db seed
```

3. **Test Accounts:** Keep this credential matrix readily available. All accounts share the password

Password@123 (unless overridden in your `.env`):

- Super Admin: `superadmin@buildorbit.dev`
- HR Admin: `hr@buildorbit.dev`
- Sales Lead: `lead@buildorbit.dev`
- Employee/Engineer: `engineer@buildorbit.dev`

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## 2. Authentication & Strict RBAC Verification

**Objective:** Prove that the system forcefully isolates user views and capabilities based on their cryptographically verified JWT roles.

1. Navigate to the login portal: `http://localhost:3000/login`

2. **Employee Isolation Test:**

- Log in as the Employee ( `engineer@buildorbit.dev` ).
- Observe the left-hand navigation sidebar. Verify that *only* the "Workforce" modules are visible (Dashboard, Attendance, Leave, Payslips).
- Manually attempt to navigate to a restricted URL by typing `http://localhost:3000/hr/payroll` into the address bar. Verify that the system actively blocks this and redirects or throws an unauthorized error.
- Log out.

3. **HR Admin Access Test:**

- Log in as the HR Admin ( `hr@buildorbit.dev` ).
  - Observe the sidebar. Verify that the "HR Management" suite (Leave Approval, Payroll, Shifts, Sites) is fully unlocked and accessible.
  - Log out.
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### 3. HR Operations: Shift Creation & Employee Mapping

**Objective:** Validate that the HR department can successfully define operational shifts and map the workforce to them dynamically.

1. Log in as the **HR Admin**.
  2. Navigate via the sidebar to `HR Management > Shifts ( /hr/shifts )`.
  3. **Create a Shift:**
    - Locate the "Create New Shift" form pane.
    - Input the following parameters:
      - Name: "Graveyard Shift"
      - Start Time: "23:00"
      - End Time: "07:00"
      - Grace Period: "15" (minutes)
    - Click the **Create Shift** button. Verify that the page refreshes and the "Graveyard Shift" appears as a distinct card in the UI.
  4. **Assign an Employee:**
    - Within the newly created "Graveyard Shift" card, locate the "Assigned Employees" section.
    - Click the dropdown menu, which dynamically populates with unassigned personnel. Select an employee (e.g., "Love Varshney").
    - Click the `+ submit` button.
    - **Verification:** Confirm that the employee's name instantly renders as a UI pill inside the shift card, proving the `assignEmployeeToShiftAction` server action successfully wrote to the `ShiftAssignment` database table.
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### 4. CRM Operations: Lead Generation & Invoicing

**Objective:** Ensure the Sales department can track prospects without data loss and convert successful deals into financial invoices.

1. Log in as the **Sales Lead** ( `lead@buildorbit.dev` ).
2. **Lead Generation (Data Integrity Test):**
  - Navigate to `CRM > Leads ( /crm/leads )`.
  - Click the **Create Lead** modal button.
  - Crucial Test: Verify the explicit presence of the **Contact Person Name** field (this was previously a documented bug).
  - Input test data:
    - Title: "Q4 Enterprise Licensing"
    - Contact Person Name: "Robert CEO"
    - Email: "[robert@enterprise.com](mailto:robert@enterprise.com)"
    - Status: "New"
  - Click **Create Lead**. Verify the lead appears in the table and that "Robert CEO" is accurately displayed in the correct column.

### 3. Invoice Generation:

- Navigate to CRM > Invoices ( /crm/invoices ).
  - Click the **Create Invoice** button (previously a disabled placeholder).
  - In the modal, select a pre-existing Client from the dropdown.
  - Enter an Invoice Number (e.g., "INV-2026-001"), a numeric Amount (e.g., "150000"), and set a Due Date in the future.
  - Click **Create Invoice**.
  - **Verification:** Confirm the invoice renders immediately as a financial record card on the screen.
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## 5. Executive Dashboard Metrics Integrity

**Objective:** Prove that executive dashboards are immune to hardcoded "dummy" data and accurately reflect backend realities.

1. Log in as the **Super Admin**.
  2. Navigate to the primary Dashboard ( /dashboard ).
  3. **Visual Verification:** Ensure the page loads smoothly without React hydration crashes.
  4. **Data Integrity Checks:**
    - **Active Users Metric:** Check the top metric cards. The "Active Users" count must exactly match the number of non-deleted, active users generated by the seed script (typically 6-7).
    - **Recent Login Activity:** Scroll to the login tracking widget. Ensure it reads recent, genuine session starts (e.g., your own login from 2 minutes ago) rather than fake placeholder data.
    - **Charts:** Review the "Workforce Trends" and "CRM Pipeline" charts. They should reflect the exact ratios of the database (e.g., a spike in attendance matching the seed script's 14-day history generation).
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## 6. HR Payroll Boundary Security Test

**Objective:** Verify that the severe React Server/Client boundary leakage bug has been permanently eradicated from the Payroll module.

1. Log in as the **HR Admin**.
  2. Navigate to HR Management > Payroll ( /hr/payroll ).
  3. **Verification:** The page must load immediately and render the current month's draft payroll data grid.
    - If the screen goes completely blank, or displays a "500 Internal Server Error" referencing `PrismaClient cannot be serialized`, the bug has regressed.
    - If the grid loads successfully and displays employee salaries marked as "Draft", the architectural fix is confirmed successful and secure.
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*End of Testing Guide. If all steps pass without deviation, the BuildOrbit platform is certified for production deployment.*